

Communication-Constrained Soft-Information Fusion for Multi-UAV OTFS-ISAC Cooperative Sensing

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Abstract—In multi-UAV OTFS-ISAC networks, inter-UAV soft-information exchange can improve cooperative sensing, but excessive signaling overhead, exchange delay, and unreliable reporting links may offset the sensing gain. To address this issue, this paper proposes a communication-error-calibrated and coarse-to-fine (C2F) delay-Doppler (DD)-aware Lagrangian link-selection method for communication-constrained multi-UAV cooperative sensing. The proposed method evaluates each target-dependent cooperative link according to its calibrated marginal deflection gain, reporting reliability, exchange delay, and DD-domain energy-capture efficiency. A coarse DD-domain Lagrangian utility is first used to screen candidate links, and local DD-grid refinement is then applied only to shortlisted candidates, thereby reducing unnecessary fine-grid evaluations. Simulation results in a 15-UAV, 10-target aerial sensing scenario show that the proposed method retains about 94.8% of the all-neighbor detection probability while reducing signaling overhead and exchange delay by approximately 92.1% and 94.8%, respectively. Under the same per-target selected-link budget, it outperforms Sensing-SINR-based selection and achieves comparable detection to Top-K Deflection with about 29% lower exchange delay.

Index Terms—Multi-UAV networks, OTFS-ISAC, cooperative sensing, soft-information fusion, DD-domain refinement, Lagrangian link selection.

I. INTRODUCTION

Integrated sensing and communication (ISAC) enables wireless networks to reuse spectrum, hardware, and signal-processing resources for both data transmission and environmental sensing [1], [2]. UAV-enabled ISAC extends this capability to aerial sensing scenarios, where platform mobility can improve coverage, line-of-sight availability, and sensing geometry [5], [6]. When multiple UAVs cooperate, the same target can be observed from different bistatic geometries, providing complementary spatial information for more reliable sensing decisions [3].

In such a distributed network, however, inter-UAV reporting links have limited rate and reliability, so more cooperative observations do not necessarily lead to higher fusion gain [3], [9]. Each observation is generated by a specific transmitter–receiver–target geometry, reported over an inter-UAV link, and

then fused with the others, and its contribution can be weakened at any stage of this sensing–reporting–fusion process. A geometrically favorable return may suffer from inefficient delay-Doppler (DD)-domain energy capture [10], [11]; its soft statistic may be distorted by unreliable reporting [7], [8]; or its reporting cost may be too large relative to its marginal fusion gain [4], [9]. As a result, all-neighbor fusion or sensing-SINR-based activation can be inefficient when inter-UAV communication resources are limited.

Existing studies have examined these aspects, though largely from separate viewpoints. Aerial sensing architectures, mobility, and resource management are the main focus of UAV-ISAC work [5], [6]. Sensing–communication tradeoffs under resource or backhaul constraints, by contrast, have been considered in network-level ISAC and cooperative sensing studies [4], [9]. On the detection side, imperfect channel knowledge and unreliable delivery have been shown to degrade detection performance [7], [8]. Separately, OTFS-based ISAC exploits DD-domain structures for high-mobility sensing and communication [10], [11]. Each line captures one facet of the problem, yet communication-constrained triplet-level scheduling that couples these facets for multi-UAV bistatic OTFS-ISAC has received limited attention.

This gap motivates a triplet-level scheduling perspective. Node-selection and resource-allocation studies act at the level of UAV nodes or links [9], and distributed-detection studies model reporting reliability without the underlying bistatic geometry [7], [8]; neither, however, exposes the unit that actually carries a cooperative observation. In multi-UAV bistatic sensing that unit is a target-dependent transmitter–receiver–target triplet, whose value depends jointly on its bistatic sensing contribution, the reliability and delay of its reporting link, and the DD-domain energy-capture efficiency of the corresponding OTFS observation [10]. Judging these factors in isolation is misleading—a triplet with high nominal sensing-SINR can still add little after fusion if its soft statistic is reported over a slow or unreliable channel, or if its DD-domain energy is poorly captured. We therefore treat each triplet as a schedulable soft-information reporting unit and score it on a calibrated sensing contribution, reporting reliability, exchange delay, and DD-

domain efficiency taken jointly, aiming to select a compact set of cooperative observations that preserves most of the sensing gain while reducing signaling overhead and reporting delay.

Our main contributions are as follows.

- We formulate communication-constrained multi-UAV OTFS-ISAC cooperative sensing as a triplet-level soft-information link-selection problem. Each candidate is characterized by its bistatic sensing contribution, reporting reliability, exchange delay, and DD-domain energy-capture efficiency.
- We develop a communication-error-calibrated marginal-deflection metric and a C2F DD-aware Lagrangian link-selection method. The method evaluates target-dependent candidates by balancing marginal fusion gain and reporting delay, while applying local DD-grid refinement only to shortlisted links.
- Simulation results in a 15-UAV, 10-target aerial sensing scenario show that the proposed method retains most of the all-neighbor detection gain with substantially lower signaling overhead and reporting delay. It also provides a more favorable detection–delay tradeoff than sensing-SINR-based selection.

The rest of this paper is organized as follows. Section II presents the multi-UAV OTFS-ISAC system model. Section III introduces the proposed C2F DD-aware Lagrangian link-selection method. Section IV provides simulation results and discussion. Section V concludes the paper.

II. SYSTEM MODEL

We consider a sensing-centric multi-UAV OTFS-ISAC network, as shown in Fig. 1(a). The UAV and target sets are denoted by $\mathcal{M} = \{1, \dots, M\}$ and $\mathcal{Q} = \{1, \dots, Q\}$, respectively. UAV i and target q have position–velocity pairs $(\mathbf{p}_i, \mathbf{v}_i)$ and $(\mathbf{p}_q, \mathbf{v}_q)$. A cooperative candidate is indexed by the target-dependent triplet (i, j, q) , where UAV i transmits an OTFS-ISAC probing signal, target q reflects it, and UAV j receives the bistatic echo. The inter-UAV reporting topology is represented by a directed graph $\mathcal{G} = (\mathcal{M}, \mathcal{E})$, where $(i, j) \in \mathcal{E}$ if $\|\mathbf{p}_i - \mathbf{p}_j\| \leq d_{\max}$ and $i \neq j$. In the scheduling model, the same ordered pair (i, j) indexes the bistatic sensing pair and the effective reporting relation associated with this candidate. We use a geometry-driven soft-information abstraction rather than a full waveform-level OTFS receiver model. The OTFS structure enters the scheduling problem through DD-bin validity, fractional DD energy loss, same-bin collision, and local DD-grid refinement. These DD-domain quantities determine whether a target-dependent triplet provides useful soft information after reporting and fusion.

A. Communication and Reporting Model

The transmit power of UAV i is split into sensing and communication parts, i.e.,

$$P_i^s = \rho P_i, \quad P_i^c = (1 - \rho) P_i. \quad (1)$$

For a directed inter-UAV link (a, b) , the communication-channel power gain is

$$g_{ab}^c = \left(\frac{\lambda}{4\pi} \right)^2 d_{ab}^{-\alpha_c} \zeta_{ab} |h_{ab}^c|^2, \quad d_{ab} = \|\mathbf{p}_a - \mathbf{p}_b\|. \quad (2)$$

Here λ , α_c , ζ_{ab} , and h_{ab}^c denote the carrier wavelength, path-loss exponent, shadowing, and small-scale fading, respectively.

The communication SINR from UAV i to UAV j is modeled as

$$\gamma_{ij}^c = \frac{P_i^c g_{ij}^c}{N_0 + \sum_{k \in \mathcal{M} \setminus \{i, j\}} (P_k^c + \epsilon_s P_k^s) g_{kj}^c + \epsilon_d P_i^s g_{ij}^c}. \quad (3)$$

The coefficients ϵ_s and ϵ_d represent neighbor-induced sensing-to-communication leakage and intra-link leakage from the sensing component of UAV i to the communication receiver, respectively.

With communication bandwidth $B_c = N\Delta f$, the reporting rate, link reliability, and exchange delay are

$$R_{ij} = B_c \log_2(1 + \gamma_{ij}^c), \quad \chi_{ij} = \frac{\gamma_{ij}^c}{\gamma_{ij}^c + \gamma_{\text{req}}}, \quad c_{ijq} = \frac{B_q}{R_{ij}}, \quad (4)$$

where $\gamma_{\text{req}} = 2^{R_{\min}/B_c} - 1$ and $B_q = K_{\text{pkt}} b_d$ is the soft-information payload size. A reporting link is feasible when $R_{ij} \geq R_{\min}$.

B. DD-Domain Bistatic Sensing Model

For triplet (i, j, q) , define $d_{iq} = \|\mathbf{p}_q - \mathbf{p}_i\|$, $d_{jq} = \|\mathbf{p}_q - \mathbf{p}_j\|$, $\mathbf{u}_{iq} = (\mathbf{p}_q - \mathbf{p}_i)/d_{iq}$, and $\mathbf{u}_{jq} = (\mathbf{p}_q - \mathbf{p}_j)/d_{jq}$. The bistatic delay and Doppler shift are

$$\tau_{ijq} = \frac{d_{iq} + d_{jq}}{c}, \quad \nu_{ijq} = \frac{(\mathbf{v}_q - \mathbf{v}_i)^T \mathbf{u}_{iq} + (\mathbf{v}_q - \mathbf{v}_j)^T \mathbf{u}_{jq}}{\lambda}. \quad (5)$$

The corresponding DD-grid indices are

$$l_{ijq} = \text{round}(\tau_{ijq} L \Delta f), \quad k_{ijq} = \text{round}(\nu_{ijq} N T). \quad (6)$$

A triplet is DD-valid if $0 \leq l_{ijq} < L$ and $-N/2 \leq k_{ijq} < N/2$. The fractional DD offsets are

$$\varepsilon_{ijq}^{\tau} = \tau_{ijq} L \Delta f - l_{ijq}, \quad \varepsilon_{ijq}^{\nu} = \nu_{ijq} N T - k_{ijq}. \quad (7)$$

Using the normalized sinc function, the coarse DD-domain gain is

$$\eta_{ijq}^c = \text{sinc}^2(\varepsilon_{ijq}^{\tau}) \text{sinc}^2(\varepsilon_{ijq}^{\nu}). \quad (8)$$

For C2F DD refinement, a local $(2W + 1) \times (2W + 1)$ DD neighborhood provides the recoverable energy η_{ijq}^{loc} , and the refined gain is

$$\eta_{ijq}^f = \min\{1, \max[\eta_{\min}, \eta_{ijq}^c + \kappa_{\text{dd}}(\eta_{ijq}^{\text{loc}} - \eta_{ijq}^c)]\}. \quad (9)$$

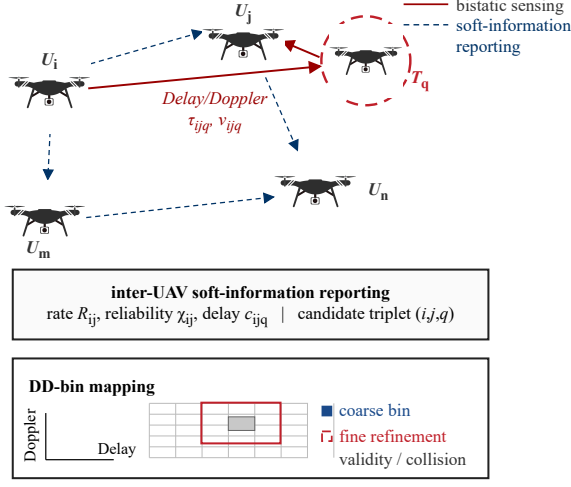
Here $\kappa_{\text{dd}} \in [0, 1]$ controls the refinement strength and $\eta_{\min}$ is the retained-energy floor.

If multiple DD-valid targets observed by the same UAV pair fall into the same DD bin, we use the collision penalty

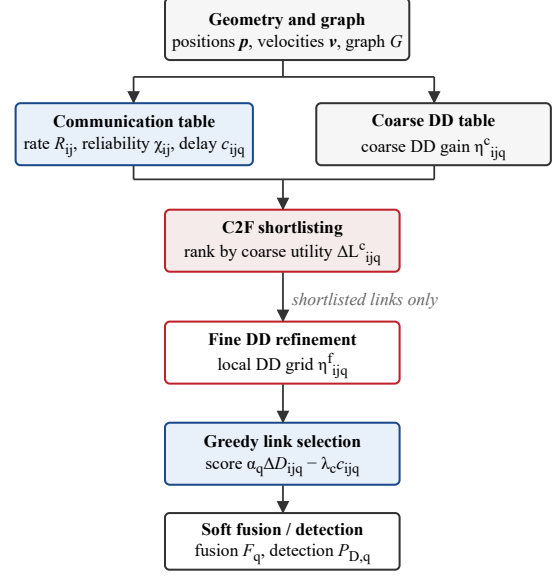
$$\eta_{ijq}^{\text{col}} = \frac{1}{n_{ijq}^{\alpha_{\text{col}}}}, \quad (10)$$

where n_{ijq} is the number of targets sharing bin (l_{ijq}, k_{ijq}) .

(a) Multi-UAV OTFS-ISAC Scenario



(b) C2F DD-Aware Selection and Fusion



Coarse table: screening; fine table: final selection, fusion, and detection.

Fig. 1. Multi-UAV OTFS-ISAC cooperative sensing scenario and C2F DD-aware link-selection framework.

The bistatic target gain is

$$g_{ijq}^s = \frac{\lambda^2 \sigma_q}{(4\pi)^3 d_{iq}^2 d_{jq}^2}, \quad (11)$$

where σ_q denotes the target RCS. Combining bistatic gain, processing gain, DD-domain gain, and collision penalty gives

$$\gamma_{ijq}^s = \frac{\tilde{P}_i^s g_{ijq}^s G_p \eta_{ijq}^{DD} \eta_{ijq}^{col}}{N_0 + I_{ij}^{res}}, \quad \eta_{ijq}^{DD} \in \{\eta_{ijq}^c, \eta_{ijq}^f\}. \quad (12)$$

In the default sensing-only mode, $\tilde{P}_i^s = P_i^s$, and $G_p = NL$. The residual term I_{ij}^{res} captures self-interference, direct leakage, and multi-UAV sensing leakage at the sensing receiver.

C. Soft-Information Fusion Model

Each feasible triplet produces a scalar soft statistic s_{ijq} . Its target-present mean and $\mathcal{H}_0$ -side variance are modeled as

$$\mu_{ijq} = \kappa_\mu \log(1 + \gamma_{ijq}^s), \quad \sigma_{ij,0}^2 = \sigma_0^2 (1 + \xi I_{ij}^{res}/N_0). \quad (13)$$

Communication-error calibration modifies the statistics used for fusion as

$$\mu_{ijq}^{eff} = \chi_{ij} \mu_{ijq}, \quad \bar{\sigma}_{ij,0}^2 = \chi_{ij} \sigma_{ij,0}^2 + (1 - \chi_{ij}) \sigma_{err}^2. \quad (14)$$

For selected set $\mathcal{S}_q$, the fused statistic is

$$F_q = \sum_{(i,j) \in \mathcal{S}_q} w_{ijq} s_{ijq}, \quad w_{ijq} \propto \frac{\mu_{ijq}^{eff}}{\bar{\sigma}_{ij,0}^2}. \quad (15)$$

The corresponding composite deflection is

$$D_q(\mathcal{S}_q) = \frac{\left(\sum_{(i,j) \in \mathcal{S}_q} w_{ijq} \mu_{ijq}^{eff} \right)^2}{\sum_{(i,j) \in \mathcal{S}_q} w_{ijq}^2 \bar{\sigma}_{ij,0}^2}. \quad (16)$$

Approximating F_q as Gaussian, the detection probability is

$$P_{D,q} \approx Q \left(Q^{-1}(P_{FA}) - \sqrt{D_q(\mathcal{S}_q)} \right), \quad (17)$$

where $Q(\cdot)$ denotes the Gaussian Q -function.

III. PROPOSED C2F DD-AWARE LAGRANGIAN LINK SELECTION

We select a compact set of cooperative candidates whose marginal fusion gain justifies their reporting cost. The selection is implemented as a low-complexity greedy rule guided by a Lagrangian marginal-gain score. For target q , the feasible candidate set is

$$\mathcal{E}_q = \{(i,j) \in \mathcal{E} : R_{ij} \geq R_{\min}, (l_{ijq}, k_{ijq}) \text{ is DD-valid}\}. \quad (18)$$

Here c_{ijq} and the DD-valid condition are defined in Section II. We use $\tilde{c}_{ijq} = 10^3 c_{ijq}$ as the delay in milliseconds.

Given the current selected set $\mathcal{S}_q$, the target-priority coefficient is updated as

$$\alpha_q = \frac{\partial P_{D,q}}{\partial D_q} + \mu_\alpha \frac{[D_{\min} - D_q(\mathcal{S}_q)]^+}{D_{\min}}, \quad (19)$$

where $[x]^+ = \max(x, 0)$. The derivative follows from (17) with a small regularization when D_q is close to zero, and $\alpha_q^{(0)}$ denotes the value evaluated at $\mathcal{S}_q = \emptyset$.

For candidate (i, j) and target q , the marginal calibrated deflection gain is

$$\Delta D_{ijq} = D_q(\mathcal{S}_q \cup \{(i, j)\}) - D_q(\mathcal{S}_q), \quad (20)$$

where $D_q(\cdot)$ uses the calibrated statistics in (14). The Lagrangian score is

$$\Delta \mathcal{L}_{ijq} = \alpha_q \Delta D_{ijq} - \lambda_c \tilde{c}_{ijq}, \quad (21)$$

where $\lambda_c > 0$ controls the delay penalty.

To reduce local DD-grid refinement cost, we use a C2F strategy. Let L_{all} be the number of feasible candidates before screening and $L_{\text{short}} \ll L_{\text{all}}$ be the shortlist size. In the coarse stage, candidates are ranked by

$$\Delta \mathcal{L}_{ijq}^c = \alpha_q^{(0)} \Delta D_{ijq}^c - \lambda_c \tilde{c}_{ijq}, \quad (22)$$

where ΔD_{ijq}^c is computed using the coarse DD gain in (8). Only the top L_{short} candidates are retained for fine refinement.

The fine stage applies (9) only to the shortlisted candidates and recomputes (20) and (21). Starting from $\mathcal{S}_q = \emptyset$, the algorithm repeatedly selects the remaining candidate with the largest positive $\Delta \mathcal{L}_{ijq}$ and updates $D_q(\mathcal{S}_q)$ and α_q . It stops when no positive-score candidate exists, the link-number limit is reached, or all active targets with feasible candidates satisfy $D_q(\mathcal{S}_q) \geq D_{\min}$. C2F reduces fine DD-grid evaluations from $\mathcal{O}(QL_{\text{all}}(2W+1)^2)$ to $\mathcal{O}(QL_{\text{short}}(2W+1)^2)$.

IV. SIMULATION RESULTS

We evaluate the proposed communication-error-calibrated and C2F DD-aware Lagrangian link-selection scheme in a high-mobility multi-UAV aerial sensing scenario. Unless otherwise specified, 15 UAVs cooperatively sense 10 aerial targets in a 4000 m $\times$ 4000 m area. The UAVs and targets move with speeds of 30–60 m/s and 50–90 m/s, respectively. We use the all-neighbor scheme as an upper-resource reference and set the target false-alarm probability to $P_{\text{FA}} = 0.05$ in all tests. Table I lists the main simulation parameters.

A. Performance and Tradeoff Analysis

For the main comparison in Fig. 2, the proposed method achieves $P_D = 0.6648$, which retains about 94.8% of the all-neighbor detection probability. At the same time, it reduces signaling overhead and exchange delay by approximately 92.1% and 94.8%, respectively. In absolute terms, the exchange delay decreases from about 5.1×10^2 ms under all-neighbor fusion to about 27 ms with the proposed method. This confirms that all-neighbor fusion is a useful upper-resource reference, but its reporting cost is too high for low-latency cooperative sensing.

A major source of this gain is the avoidance of ineffective soft-information reporting. Sensing-SINR-based selection tends to choose links with strong sensing responses, without explicitly checking whether their soft statistics can be reported reliably and quickly. By combining communication-error-

TABLE I
MAIN SIMULATION PARAMETERS

Parameter	Value
UAVs / targets	$M = 15, Q = 10$
Area and comm. range	$4000 \times 4000 \text{ m}^2, d_{\max} = 2500 \text{ m}$
UAV / target speed	$30\text{--}60 \text{ m/s} / 50\text{--}90 \text{ m/s}$
Carrier and DD grid	$f_c = 5.9 \text{ GHz}, N = L = 64, \Delta f = 30 \text{ kHz}$
Power split / Tx power	$\rho = 0.8, P_i = 1 \text{ W}$
Comm. channel	$\alpha_c = 2$, shadowing std. 1 dB, Rician $K = 15 \text{ dB}$
Rate constraint / false alarm	$R_{\min} = 0.2 \text{ Mbit/s}, P_{\text{FA}} = 0.05$
Payload size	$K_{\text{pkt}} = 4, b_d = 160 \text{ bits}, B_q = 640 \text{ bits}$
Link-selection parameters	$D_{\min} = 3, \lambda_c = 0.005, L_{\max} = 6, L_{\text{tot}} = 60$
C2F DD parameters	$W = 1, \kappa_{\text{dd}} = 0.45, \eta_{\min} = 0.25, L_{\text{short}} = 25$
Monte Carlo trials	1000

calibrated marginal deflection with delay-aware Lagrangian pricing, the proposed method keeps most useful cooperative observations and discards links whose reporting cost is not justified by their effective fusion contribution.

Under the same per-target selected-link budget, the proposed method achieves higher detection probability than Sensing-SINR and Top-K Sensing-SINR. Compared with Top-K Deflection, it provides similar detection performance while reducing exchange delay by about 29%. This behavior is consistent with the delay-aware Lagrangian term, which discourages unreliable or slow soft-information links even when their sensing or deflection scores are high.

Varying λ_c in Fig. 3 changes the selected-link pattern directly. A larger λ_c reduces the number of selected links, signaling overhead, and exchange delay, with only a moderate loss in detection probability. Hence, λ_c serves as a practical control knob for the detection–delay tradeoff. Changing $R_{\min}$ in Fig. 4 affects the problem in a different way: a larger rate requirement shrinks the feasible reporting graph and weakens the cooperative sensing gain. In the low-to-moderate constraint region, the proposed method still maintains a better sensing–communication tradeoff than Sensing-SINR by prioritizing reliable and low-delay links.

Taken together, these two sensitivity tests separate the roles of algorithmic pricing and communication feasibility. The Lagrangian price adjusts the penalty on reporting delay, whereas the rate constraint changes which inter-UAV links can be used for soft-information exchange. When the feasible reporting graph becomes too sparse, all cooperative schemes degrade, indicating that communication feasibility remains a fundamental bottleneck.

B. Ablation and Robustness Analysis

For the component study in Table II, all variants are evaluated under a fixed selected-link budget. The “Full” row is therefore used for within-table comparison and is not intended to duplicate the main-comparison value in Fig. 2. Among the tested variants, removing communication-error-

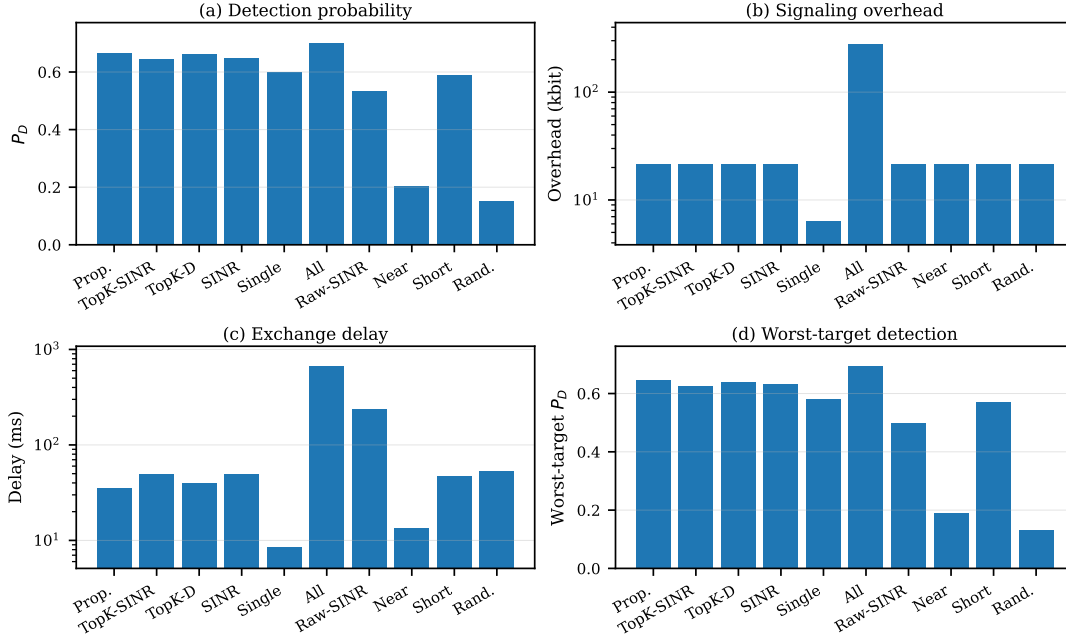


Fig. 2. Main performance comparison under C2F DD refinement.

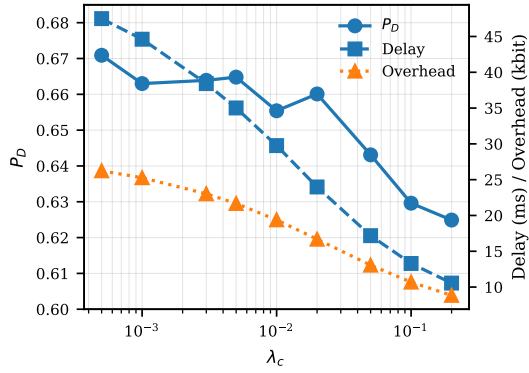


Fig. 3. Sensitivity to the Lagrangian communication price λ_c .

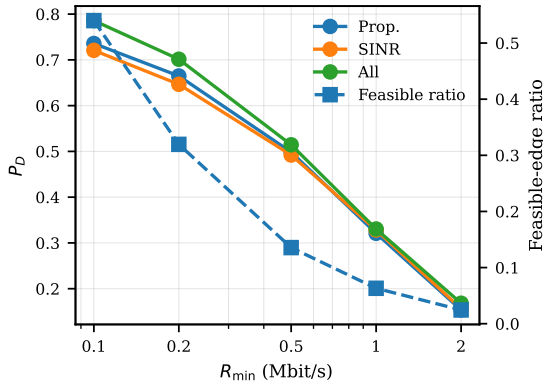


Fig. 4. Sensitivity to the communication-rate constraint $R_{\min}$.

TABLE II
COMPONENT ABLATION UNDER THE SAME SELECTED-LINK BUDGET

Variant	P_D	Bits (kbit)	Delay (ms)	Worst P_D
Full	0.6540	16.26	26.82	0.631
w/o target priority	0.6198	16.63	29.59	0.601
w/o delay price	0.6564	16.48	29.21	0.635
w/o comm.-error calib.	0.5856	13.77	28.48	0.572
w/o C2F refine	0.6223	16.41	27.42	0.604

calibration causes the largest degradation, reducing P_D from 0.6540 to 0.5856. This highlights the importance of calibrating unreliable exchanged soft information before fusion. Without C2F refinement, P_D decreases to 0.6223, which indicates that local DD refinement helps recover fractional delay-Doppler energy leakage. Removing target priority degrades both average and worst-target detection performance. Without delay-aware pricing, P_D and worst-target P_D slightly increase, but the exchange delay also becomes larger; this matches the role of the delay-price term in trading a small amount of detection gain for lower reporting latency.

In a separate DD-domain ablation under the main-comparison configuration, C2F refinement improves P_D from 0.6262 to 0.6648. It matches full local refinement while reducing fine DD-grid evaluations by about 41.1%. This experiment isolates the effect of DD-domain refinement and shows that local DD-grid recovery is useful, whereas applying it to every feasible triplet is unnecessary.

Robustness is further evaluated under communication errors and residual interference in Fig. 5. As the soft-error scale or residual factor increases, all methods degrade. The proposed

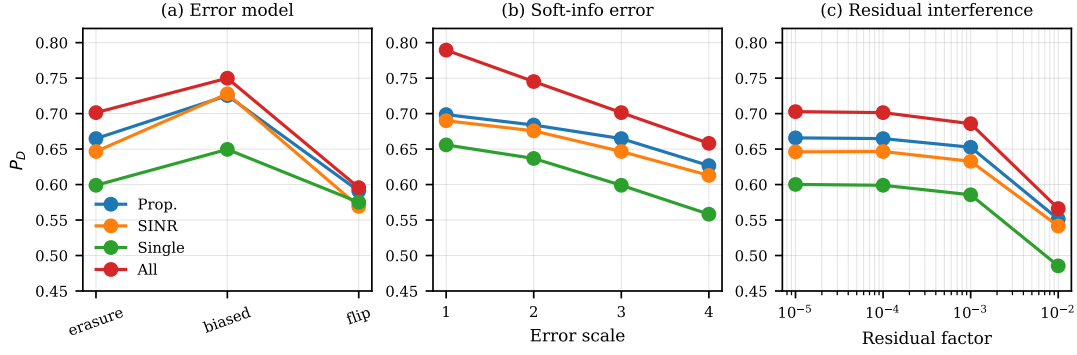


Fig. 5. Robustness under communication errors and residual interference.

method still maintains a favorable detection–delay tradeoff because communication reliability and residual-interference-aware calibration penalize unreliable links before fusion. These results support the robustness of the proposed soft-information fusion strategy under imperfect reporting conditions.

V. CONCLUSION

This paper investigated communication-constrained cooperative sensing in a multi-UAV OTFS-ISAC network. We developed a communication-error-calibrated and C2F DD-aware Lagrangian link-selection method that jointly accounts for marginal deflection gain, reporting reliability, exchange delay, and DD-domain energy-capture loss. By calibrating exchanged soft statistics according to reporting reliability, the proposed method reduces the impact of unreliable reports on the fused statistic. The C2F DD-aware design further applies local DD-grid refinement only to shortlisted candidates, thereby recovering useful DD-domain energy while avoiding unnecessary fine-grid evaluations.

Simulation results showed that the proposed method retains most of the all-neighbor detection gain with much lower signaling overhead and exchange delay. Under the same per-target selected-link budget, it achieves higher detection probability than Sensing-SINR-based selection and provides detection performance comparable to the Top-K Deflection baseline with lower delay. Ablation and robustness results further support the benefits of communication-error calibration, target-priority weighting, delay-aware Lagrangian pricing, and C2F DD refinement.

Future work will extend the current geometry-driven soft-information abstraction toward waveform-level OTFS-ISAC receiver modeling, more realistic inter-UAV reporting protocols, and joint detection-tracking design for dynamic multi-target UAV sensing networks.

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